



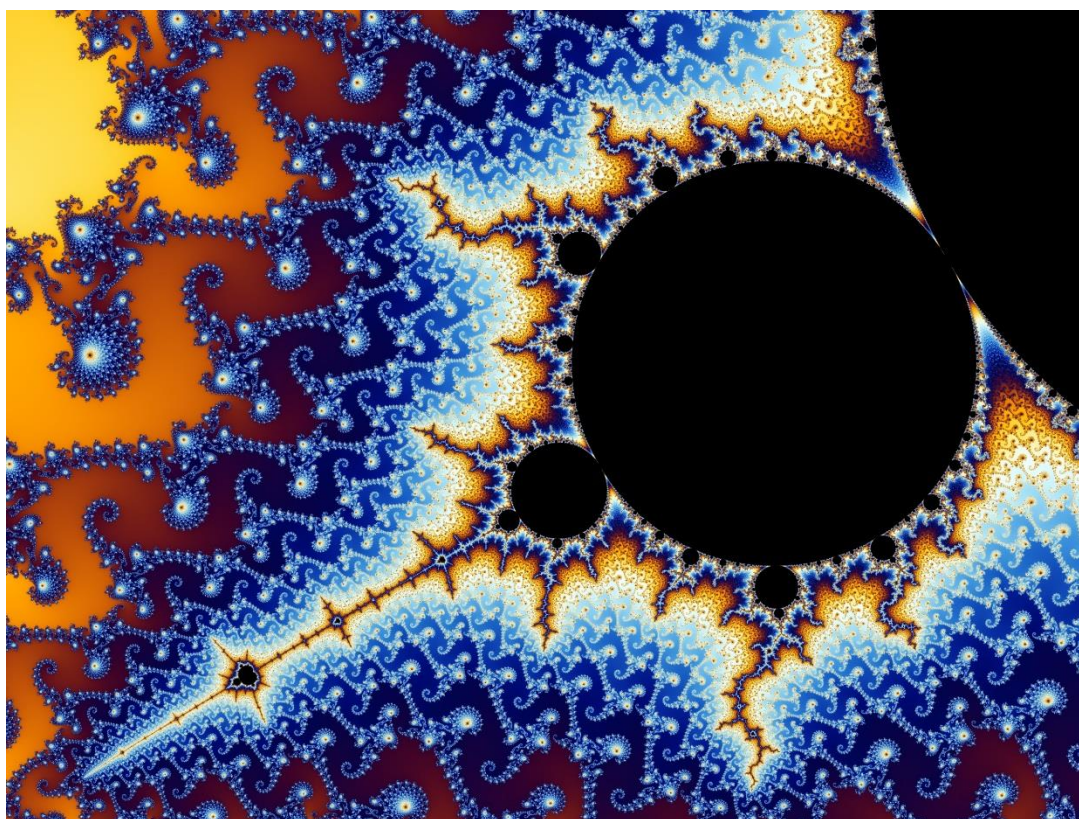
ST STITHIANS
COLLEGE



Tour de Maths 2025

Leg 1

Question Paper



The Mandelbrot set

5 Mark Questions

Question 1

In the following equation, $\sqrt{CASIO} = 304$, each letter represents a unique digit from the digits 0-9. What is the digit value of the letter S?

Question 2

How many decimals are there after the decimal point when we calculate $\frac{2025}{2^{16}}$?

Note: your calculator truncates this answer so don't trust it entirely

Question 3

What is the last digit if we expand 3^{2025} ?

Question 4

$$2025 = 1^n + 2^n + 3^n + \dots + 8^n + 9^n$$

Determine n.

Question 5

How many perfect square factors does 2025 have?

Question 6

What is the value of x in $\sqrt[3]{2025 \times x}$ that will result in an output with the smallest integer value?

Question 7

The sum of the digits of 2025 is a perfect square. Which is the next calendar year that has this property?

Question 8

Determine the value of this power tower, where each digit is the exponent of the digit before itself.

$$2^{4^{2025}} + 6^{2025} + 3^{3^{2025}} = ?$$

Question 9

$$\frac{2025}{(1+2+3+\dots+n)^p} = 1$$

Determine $n + p$

Question 10

Would you rather have 20% of a pizza slice with radius 25cm, or would you choose 25% of a pizza slice with radius 20cm?

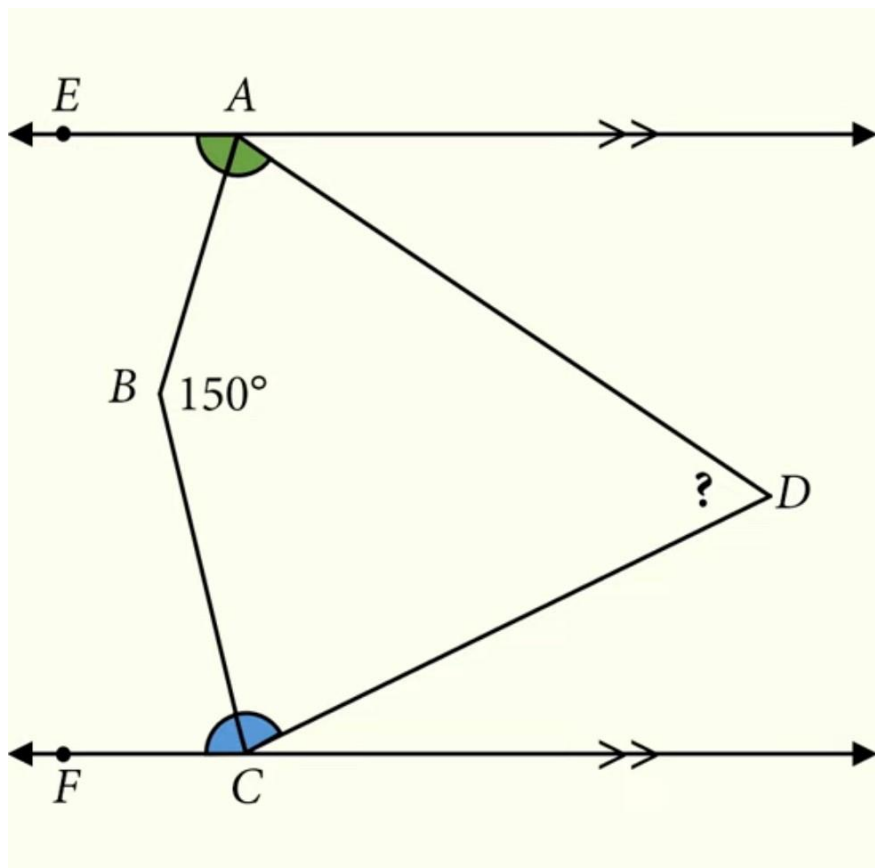
Explain your answer.



10 Mark Questions

Question 11

Determine the magnitude of D in the diagram below.



Question 12

Question 13

Question 14

Question 15

Question 16

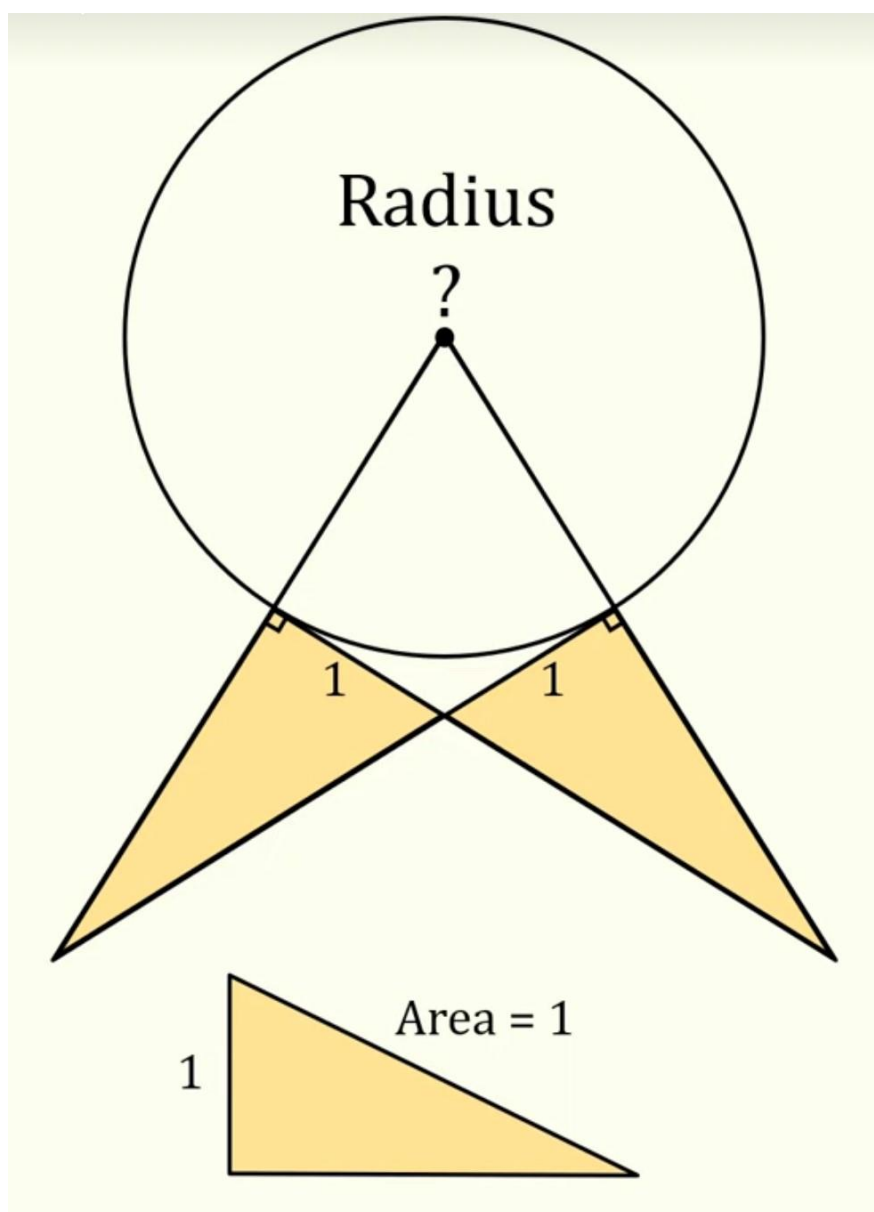
Question 17

Question 18

Question 19

20 Mark Questions

Question 20



Question 21

Question 22

Total: 200

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